Artificial Intelligence

VICTORIA UNIVERSITY OF WELLINGTON
TE HERENGA WAKA

AIML320/420

Data Preparation and Feature Engineering - 1

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AIML320/420 Week 3

1. Announcements
   - Assignment 1 (**15%**)
     Due on 11:59pm,
     26th March (Thur), week 5
   - Mid-term test, 23/04/2026, 12:00-12:50pm
   - Discord, link
   - Fangfang's teaching evaluation by 20th March, next Friday (start today)
2. Algorithms Learned:
   - K nearest neighbour (KNN)
   - Decision Tree
   - Ensemble Learning
3. Data Preparation and Feature Engineering

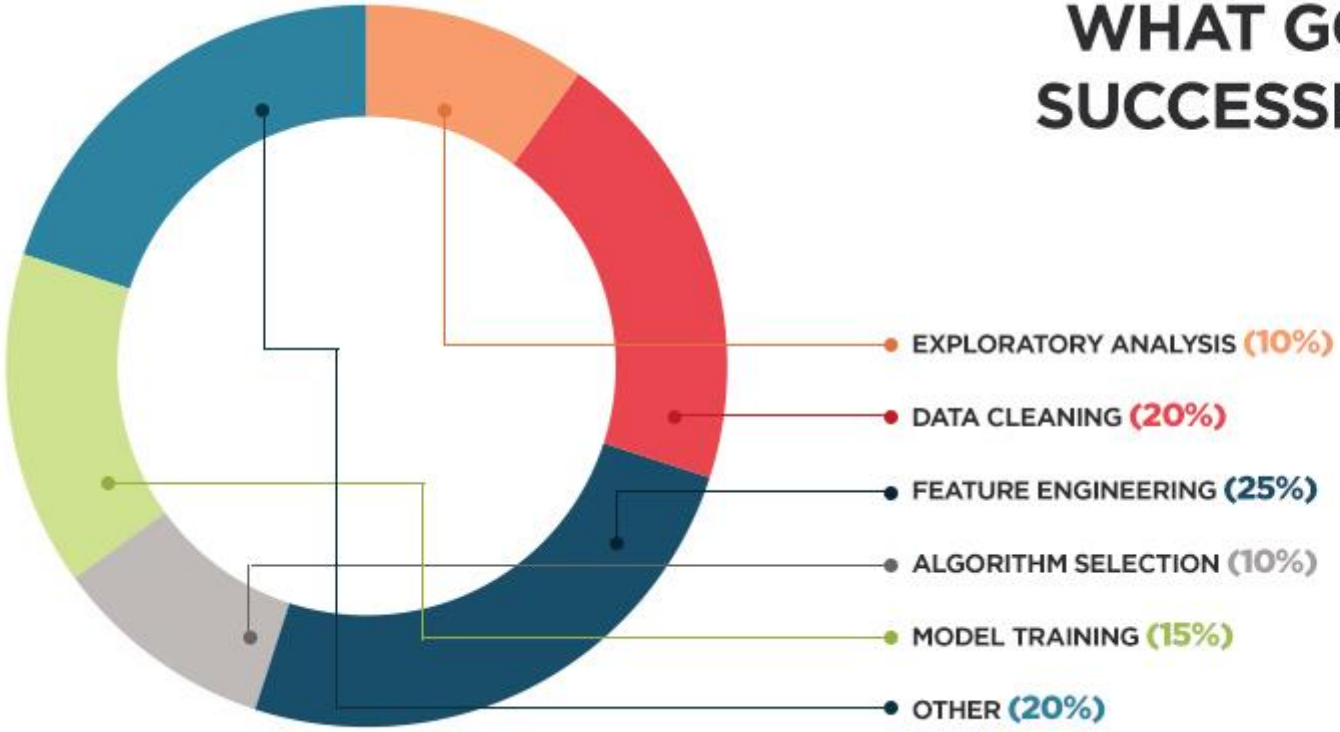
Why Data Preparation Matters

- Imagine you're about to make a healthy fruit smoothie for breakfast. What do you need to do before putting everything in the blender?
  - Wash the fruits
  - Peel bananas or oranges
  - Remove pits from cherries
  - Cut fruits into smaller pieces

- The same for a machine learning model's performance.
  - "data preparation", prepare the final quality data set(s) for modelling
  - Quality decisions must be based on quality data

~~Not rubbish in, rubbish out~~

Why Data Preparation Matters



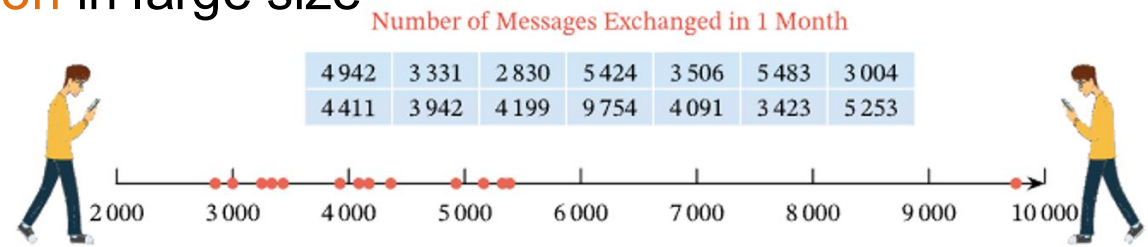
DATA SCIENCE PRIMER
by EliteDataScience

Outline

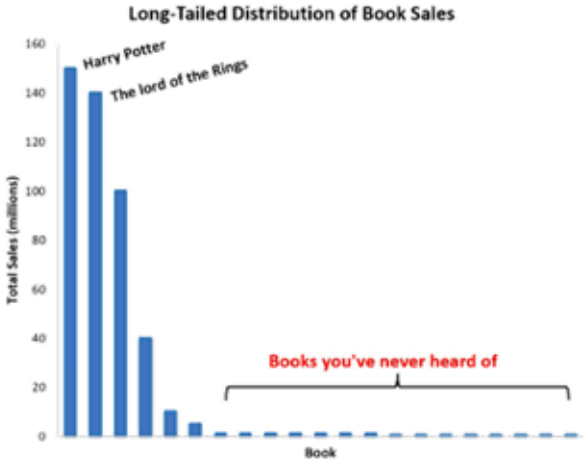
- Introduction of Data Preparation/Preprocessing
  - What data preparation include
  - Different types of data preparation techniques
    - Categorical Data Encoding
    - Data Discretisation
    - Data Normalisation
    - Data Cleaning
    - Missing data
    - Imbalanced data

Data in the Real World

- Data Type Diversity:
  - Continuous data
  - Discrete data: categorical/nominal; ordinal; Integer
  - Other/special types of data (multi-media data): Text data, hyperlink data, image data
- Unconstrained distribution in large size
  - E.g., outliers



  - Long-tail distribution:
    - E.g., book sales,



Data in the Real World

- Messy: Lots of potentially incorrect data, caused by human or computer error, transmission error
  - incomplete: lacking attribute values
  - noisy: containing noise, errors, or outliers
    - e.g., *Salary*="−10" (an error)
  - inconsistent: containing discrepancies in codes or names, e.g.,
    - *Birthday*="03/07/2010"
    - Was rating "1, 2, 3", now rating "A, B, C"
  - Intentional (e.g., *disguised missing* data)
    - Jan. 1 as everyone's birthday?

Data Preprocessing/Preparation

- Common tasks for data preprocessing:
  - Data regularization: encoding/discretization
  - Data transforms: change the scale/distribution of variables
  - Data cleaning: identify, and correct mistakes
    - Handle missing data, noisy data, outliers, redundant data
  - Dimensionality reduction: create compact projections of data
  - Feature selection: identify relevant input variables

Categorial Data Encoding Scheme

- Convert categorical data to numeric value
- Categorical variables: contain label values rather than numeric values
- One Hot Encoding: for each unique value in a categorical column, a new column is added

dummy variables

Country
USA
UK
USA
France
USA
UK

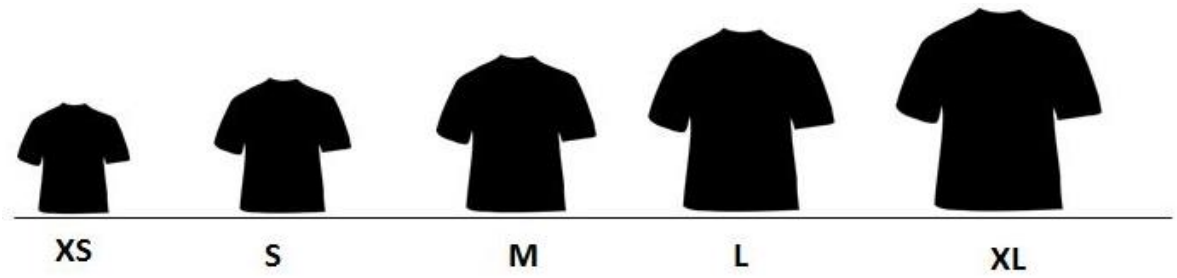
USA	UK	France
1	0	0
0	1	0
1	0	0
0	0	1
1	0	0
0	1	0

UK	France
0	0
1	0
0	0
0	1
0	0
1	0

- Dummy Encoding: N-1 columns for N unique values
  - We **do not** create a column for USA. Why?
  - Because if UK is 0 and France is 0, the model can logically deduce that the city *must* be USA.

Encoding Ordinal Variables

- Ordinal data: categorical data that have a natural rank order, e.g. *Poor, Good, Very Good, and Excellent*

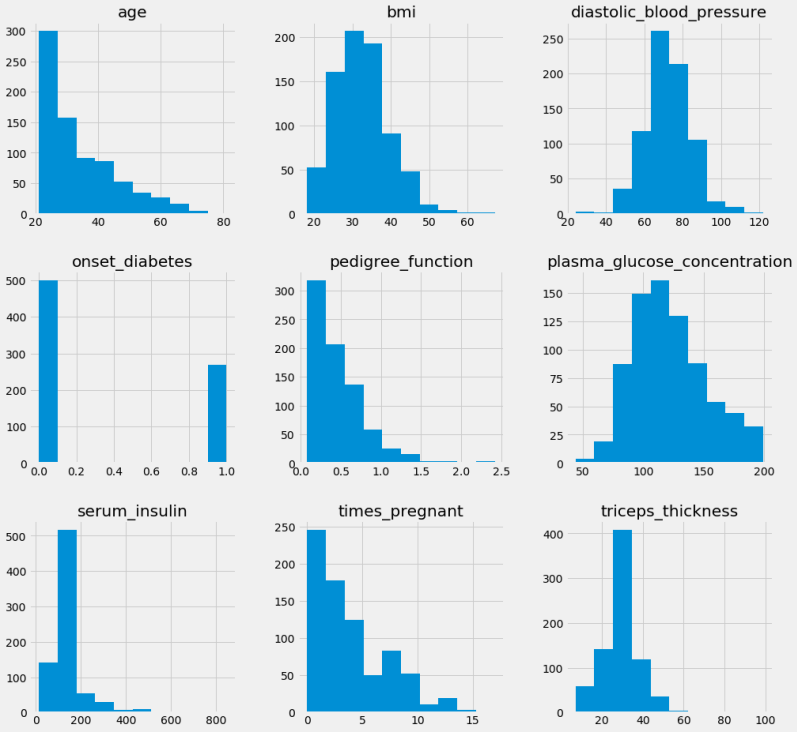


- Ordinal encoding: assign integers to labels in certain order

Original Encoding	Ordinal Encoding
Poor	1
Good	2
Very Good	3
Excellent	4

Normalisation/Scaling

- Numeric data, feature values in different ranges
- Some machine learning methods e.g. KNN, SVM, gradient descent, are affected greatly by the scale of the data, but DT are not!
- Normalization transforms data to a consistent set of rules e.g.[0, 1]



Example of variables with vastly different scales

Min-Max Normalisation/Scaling

Linear scaling

- To the range [0, 1]:

$$x' = \frac{x - X_{min}}{X_{max} - X_{min}}$$

  the min are all zeros and the max values are all ones

  Remember KNN?

- To a pre-defined range [New_{min}, New_{max}]:

$$x' = \frac{x - X_{min}}{X_{max} - X_{min}}(New_{max} - New_{min}) + New_{min}$$

- Use the scikit-learn object MinMaxScaler

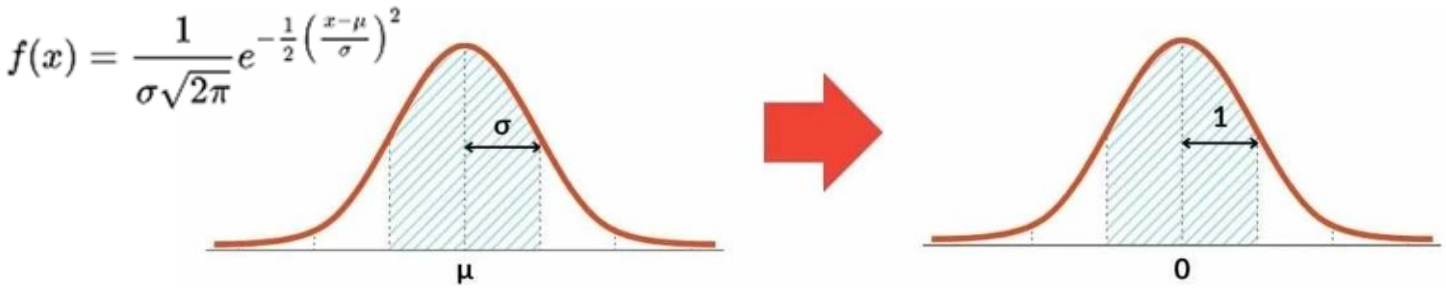
Z-Score Standardisation

- Referred to as center scaling
- Center values around a mean of zero and a standard deviation of one utilising the statistical idea: z-score/standard score

$$z = (x - \mu)/\sigma$$

μ is the mean, σ is the standard deviation of the feature

- Use the scikit-learn object StandardScaler



Standard Normal Distribution Z-Score Standardisation

e.g., the height of students aged from 8-18，Blood pressure (in a healthy population)

Normalisation or Standardisation?

- The main difference:
  - Standardisation can give values that are both positive and negative centered around zero
  - Normalisation makes different variables to have the same range

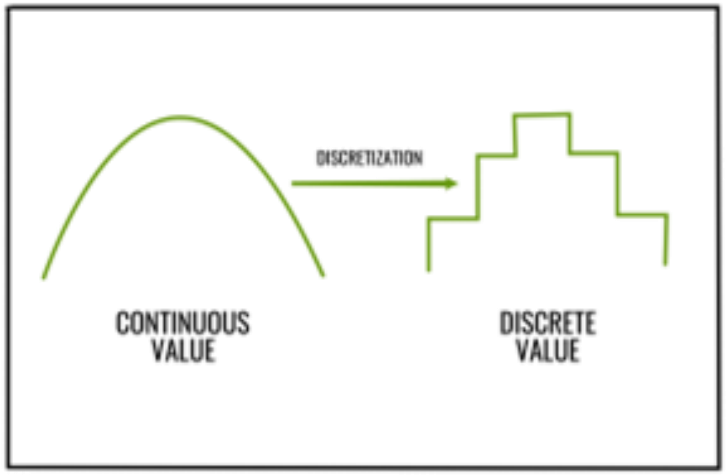
- How to choose?
  - If the distribution is normal, then it should be standardised, otherwise => normalise
  - If in doubt => normalise

- Normally make statistics on training dataset, and apply to the test dataset

- Might be a good idea to have a mixture of standardised and normalised variables=> standardised followed by normalised

Discretisation

- Discretisation/binning:
  - Converting continuous numeric values such as price, age, and weight into discrete intervals
- Why discretisation:
  - Some algorithms prefer/require discrete inputs
    - e.g. DT, rule-based algorithms
  - data smoothing, handle outliers



Age -> Age groups

Temperature -> Cold, Warm, Hot

- Use scikit-learn object KBinsDiscretizer

Discretisation: Equal-Width/Uniform

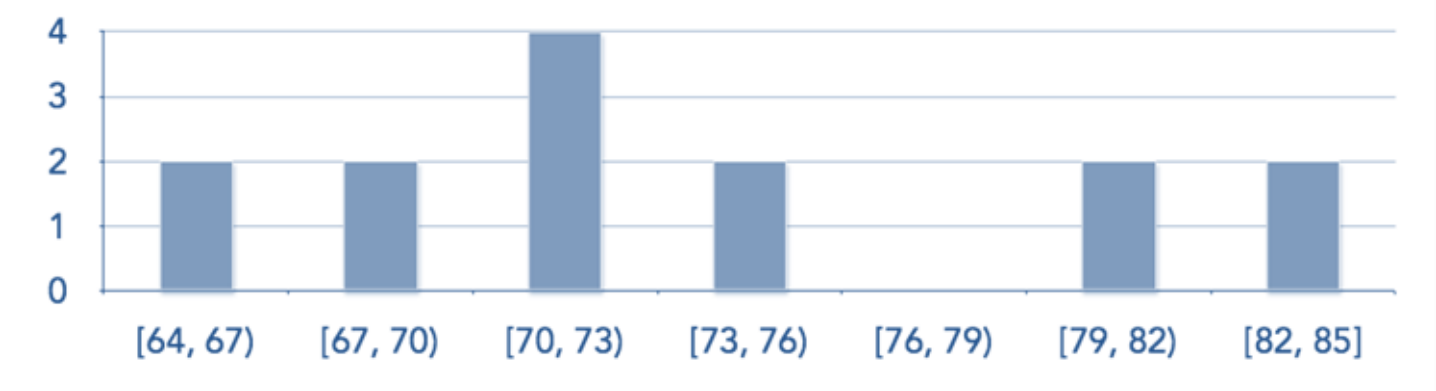
Convert a numeric attribute to a nominal/ordinal attribute with N possible values

- Find the Maximum and Minimum values of the attribute
- Divides the range [Min, Max] into N intervals of equal size
- The width of intervals: W=(Max - Min)/N

- KBinsDiscretizer(n_bin=7, encode='ordinal', strategy='uniform')

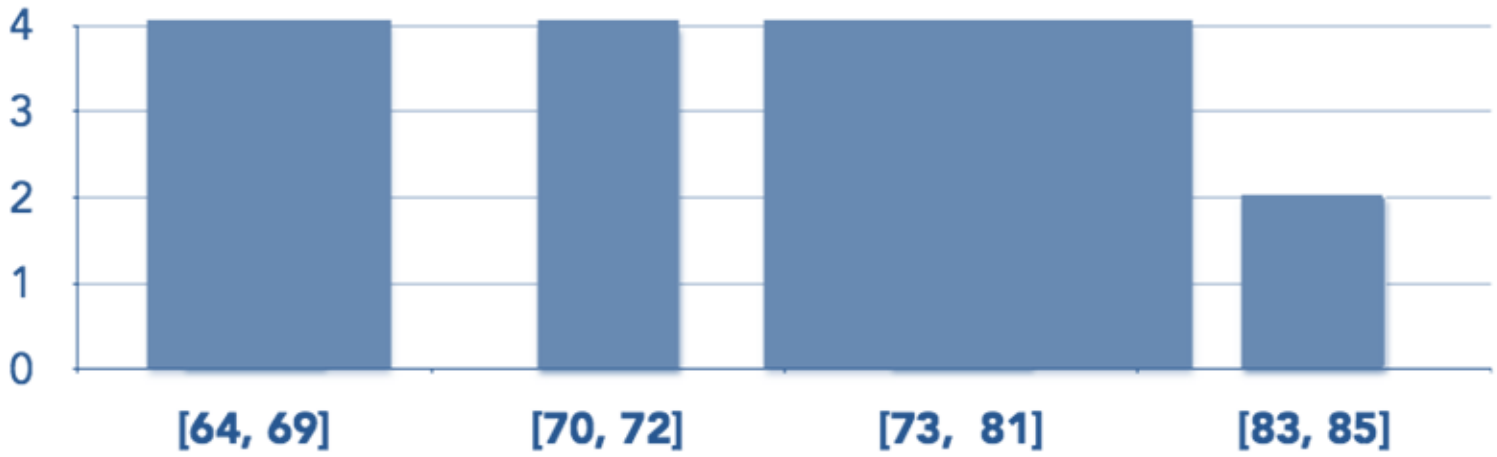
  Example: Temperature values are 85, 80, 83, 70, 64, 65, 68, 71, 69, 72, 75, 75, 81,72

  - Max=85, Min=64, N=7, W = (85-64)/7=3



Discretisation: Equal-Depth/Frequency/Quantile

- Divides the range [Max, Min] into N intervals
- Each interval including approximately same number of instances
- KBinsDiscretizer(n_bin=4, encode='ordinal', strategy='quantile')
- Example:
  - Sort the 14 Temperature values
  - 64, 65, 68, 69, 70, 71, 72, 72, 75, 75, 80, 81, 83, 85
  - N=4



Missing Data

- Values for one or more variables are missing from recorded observations
- Missing data is a common issue in almost every real dataset

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	?	No
2	? NaN	?	100K	No
3	No	Single	70K	No
4	Yes	Married	?	No
5	No	?	95K	Yes
6	?	Married	60K	No
7	Yes	Divorced	220K	No
8	No	?	?	Yes
9	?	Married	75K	No
10	No	Single	90K	Yes

Types of Missing Data

Three types of missing data:

Missing completely at random (MCAR)

- missing is unrelated to the variable of interests and other variables
  - e.g. survey responses are missing due to occasional data entry errors unrelated to respondents or survey questions

Missing at random (MAR)

- missing depends on other observed variables but not on the value of the missing data itself
  - e.g. if the likelihood of missing income data in a survey depends on the respondent's education level, but not on the actual income itself
  - If we know the values of other variables, we can predict the likelihood of the data being missing.

Missing not at random (MNAR)

- missing depends on both other observed variables and the missing data itself
  - e.g. high-income individuals are less likely to disclose their income, missingness is higher for individuals with higher actual incomes

Handling Missing Values

- Deletion approaches
  - Omits all records containing missing values. Only applies:
    - Missing data introduced in the MCAR mode,
    - When data contains less than 5% of missing values
- Imputation (estimation) approaches
  - Fill missing values with plausible values
    - Mean/Mode imputation
    - KNN imputation

Delete Incomplete Data Observations

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	?	Married	?	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	?	Divorced	?	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	?	?	75K	No
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Delete Data Attributes with Missing Values

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Imputation Approaches

- Mean imputation: for continuous attributes
  - Fills with average complete values
  - sklearn.impute.SimpleImputer (strategy='mean’)
- Mode imputation: for categorical attributes
  - Fills with the most frequent value
  - sklearn.impute.SimpleImputer (strategy=‘most_frequent’)

Estimate Missing Values

- Most common/mean value

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	No	**Single**	**105K**	

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- Mode imputation: for categorical attributes
  - Fills with the most frequent value
  - sklearn.impute.SimpleImputer (strategy=‘most_frequent’)
- KNN imputation
  - Find K nearest neighbours using observed values
  - Estimate the missing value by the mean/mode from the K neighbours
  - sklearn.impute.KNNImputer()

KNN Imputation

- KNN based on other attributes

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K-NN

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- Select the K most similar samples (neighbors) based on available features.
- Impute Missing Values based on K neighbors with Mean/mode method, i.e., mode refers to the most frequently value

Imputation Approaches

- Mean/Mode imputation:
  - Simple, fast
  - Doesn't change the mean/mode of attributes
  - Loss of information, depends on data types
- KNN imputation
  - Capture complex relationship
  - Flexible
  - High computational complexity, Parameters

Summary

- Data preprocessing is an important step
- Encoding categorical data
- Data normalisation/standardisation
- Data discretisation
- Missing data handling

Next:

- Tutorial